

Path 3 Project Report

Team Information

Team Member 1: Victor Tishkevich

Purdue Username: vtishkev

Path Taken: Path 3

GitHub Repository Link: https://github.com/Victor36564/vtishkev_Mini-Project

1. Dataset Description

Working with the digits dataset from sklearn (`sklearn.datasets.load_digits`) which is comprised of 10 classes (digits 0-9) with each datapoint being an 8 x 8 image of a digit.

2. Analysis Methodology

Analysis Question 7: Discuss your results. Is there a difference between the performance of the three models?

Method: Three different models were created and fitted to the digits data. Section 3 will have discussion on the accuracy of each of the three models.

Analysis Question 8: Describe what is happening to the training data.

Method: Now, the training data previously used for fitting the three models is poisoned. Section 3 will have discussion on how this data is getting poisoned

Analysis Question 10: Discuss how the three model performances have changed after poisoning the training data.

Method: The three models are trained using the poisoned training data. Section 3 will have discussion on the accuracy of each of the three models.

Analysis Question 11: Discuss what model showed strongest robustness against the poisoning.

Method: The models are trained using the poisoned training data. Section 3 will have discussion on how the models held up against the poisoned training data.

Analysis Question 12: Describe what is happening to remove the noise.

Method: Now, the poisoned training data is denoised. Section 3 will have discussion on how this data is being denoised using the KernelPCA method.

Analysis Question 13: Discuss how Poison Data 1 differs from the denoised data.

Method: Since the poisoned data is denoised, they will differ from each other. Section 3 will have discussion on how the poisoned data differs from the denoised data.

Analysis Question 15: Discuss how the three model performances have changed after applying the denoising steps. Have the performances improved? Or was there no significant difference?

Method: The three models are trained using the denoised training data. Section 3 will have discussion on the accuracy of each of the three models and the differences in performance between the models that were trained using poisoned data.

3. Results and Interpretation

Analysis Question 8: Discuss your results. Is there a difference between the performance of the three models?

After training the three models, Gaussian, K-NeighborsClassifier, and MLPClassifier, were trained using the training data, the overall accuracy of each model was determined against the test data. It is seen that the K-NeighborsClassifier model performed the best with a 95.46% accuracy, the MLPClassifier model performed the second best with a 91.47% accuracy, and the Gaussian model performed the worst with a 80.07% accuracy. Below is the printout of the executed code:

```
The overall results of the Gaussian model is 0.8007414272474513
The overall results of the K-NeighborsClassifier model is 0.9545875810936052
The overall results of the MLPClassifier model is 0.9147358665430955
```

Analysis Question 8: Describe what is happening to the training data.

The data is being poisoned by a random amount of noise that is getting picked from a normal distribution of values for noise and stored into the same sized array as X_train, then this array is added element wise to the X_train data to create poisoned train data. Below is the code snippet where this occurs:

```
noise_scale = 10.0
poison = rng.normal(scale=noise_scale, size=X_train.shape)
X_train_poison = X_train + poison
```

Analysis Question 10: Discuss how the three model performances have changed after poisoning the training data.

After the three models, Gaussian, K-NeighborsClassifier, and MLPClassifier, were trained using the poisoned training data, the overall accuracy of each model has changed. It is seen that the Gaussian model performed the best with an 81.46% accuracy (+1.39% change), the MLPClassifier model performed the second best with a 79.43% accuracy (-12.04% change), and the K-NeighborsClassifier model performed the worst with a 60.52% accuracy (-34.94% change). Below is the printout of the executed code:

```
The poisoned results of the Gaussian model is 0.8146431881371641
The poisoned results of the NeighborsClassifier model is 0.6051899907321594
The poisoned results of the MLPClassifier model is 0.794253938832252
```

Analysis Question 11: Discuss what model showed strongest robustness against the poisoning.

With the three models having different overall accuracies when compared to the unpoisoned training data, there can be a definition on which models showed the strongest robustness to the poisoned data. The model that had the highest robustness was the Gaussian model, as it had an increase in its overall accuracy of 1.39% after training on the poisoned data. The next most robust model was the MLPClassifier model, as it had a decrease in its overall accuracy of 12.04%, but still a relatively high overall accuracy of 79.43%. Finally, the least robust model was the K-NeighborsClassifier model, as it had a decrease in its overall accuracy of 34.94%, which is very significant considering it initially had the greatest accuracy of the three models.

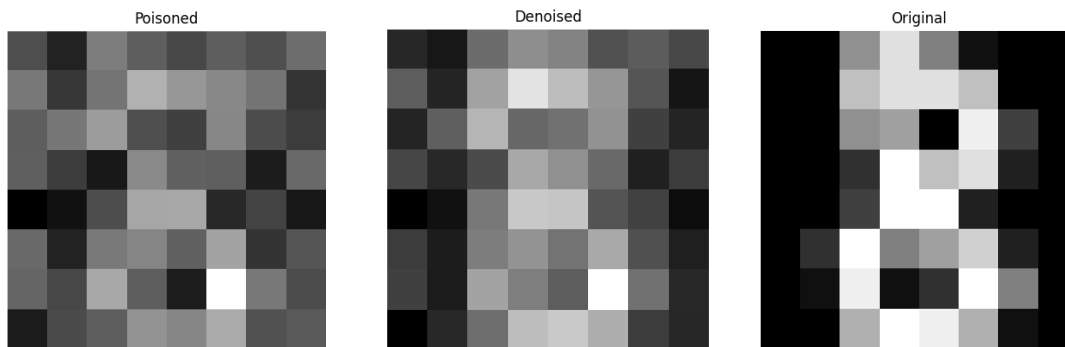
Analysis Question 12: Describe what is happening to remove the noise.

To remove the noise from the poisoned data set, the kernelPCA method uses the approximation function learned during the fitting process, filters out the low-variance data, and reconstructs an image based on the remained cleaned-up data.

Analysis Question 13: Discuss how Poison Data 1 differs from the denoised data.

The poisoned data differs from the denoised data as the digit is much more washed out in the poisoned image versus the denoised image, this effect can be seen in the images displayed below. The amount that the images differ depend on the parameters used in the kernelPCA method. In this code, the parameters used in the kernelPCA method were kept the same as the ones listed in the documentation link provided (https://scikit-learn.org/stable/auto_examples/applications/plot_digits_denoising.html). Below is the

printed images of the 8 digit from the poisoned, denoised, and original training data sets, along with the code snippet of the parameters for the kernelPCA method:



```
kernel_pca = KernelPCA(  
    n_components=400,  
    kernel="rbf",  
    gamma=1e-3,  
    fit_inverse_transform=True,  
    alpha=5e-3,  
    random_state=42,  
)
```

Analysis Question 15: Discuss how the three model performances have changed after applying the denoising steps. Have the performances improved? Or was there no significant difference?

After the three models, Gaussian, K-NeighborsClassifier, and MLPClassifier, were trained using the denoised training data, the overall accuracy of each model has changed. It is seen that the Gaussian model performed the best with an 81.84% accuracy (+0.38% change), the K-NeighborsClassifier model performed the second best with an 81.65% accuracy (+21.13% change), and the MLPClassifier model performed the worst with an 81.37% accuracy (+1.94% change). All of the models' performances have improved when using the denoised data to train. Although, the Gaussian and MLPClassifier models only improved by a small amount, less than 2%. On the other hand, since the K-NeighborsClassifier model performed very poorly when the training data was poisoned, it had a substantial increase when the poisoned data was denoised. Below is the printout of the executed code:

```
The denoised results of the Gaussian model is 0.8183503243744208  
The denoised results of the K-NeighborsClassifier model is 0.8164967562557924  
The denoised results of the MLPClassifier model is 0.8137164040778498
```